



Transformed objects

Task A: Which transformation?

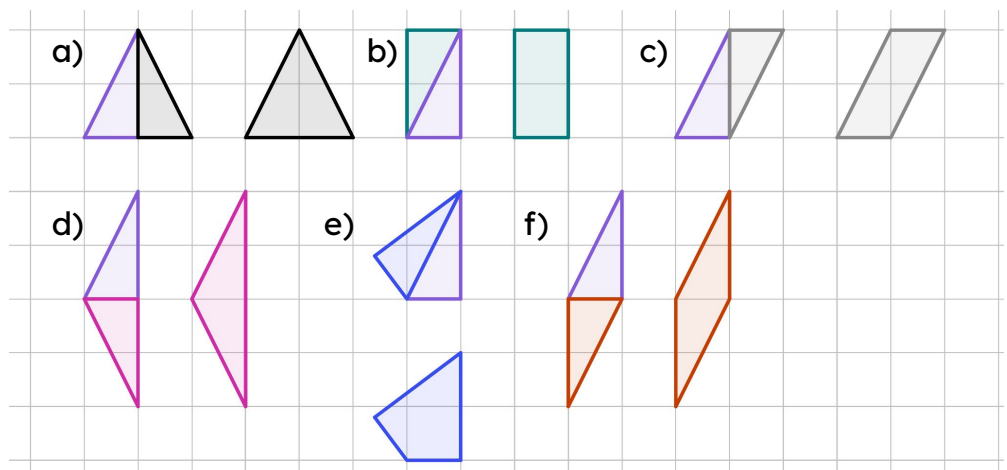
- 1) By considering the changes, decide which transformation could have taken place in each case. There might be more than one.

- a) the lengths have changed _____
- b) the sense has changed _____
- c) the orientation has changed _____
- d) the position has changed _____

- 2) By considering the invariant properties, decide which transformation could not have taken place in each case. There might be more than one.

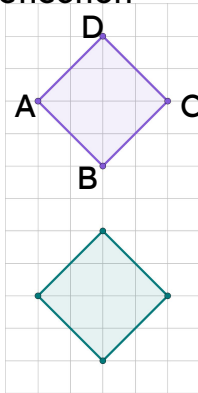
- a) the lengths are invariant _____
- b) the sense is invariant _____
- c) the orientation is invariant _____
- d) the angles are invariant _____

- 3) Identify the type of transformation that has taken place to create the new shape.

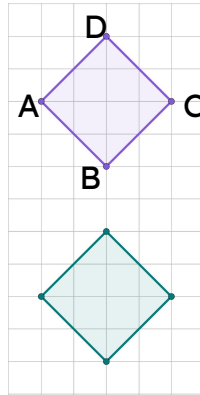


4) Add the image vertices labels.

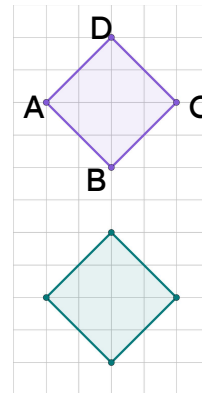
a) After a vertical reflection



b) After a translation



c) After a 180° rotation



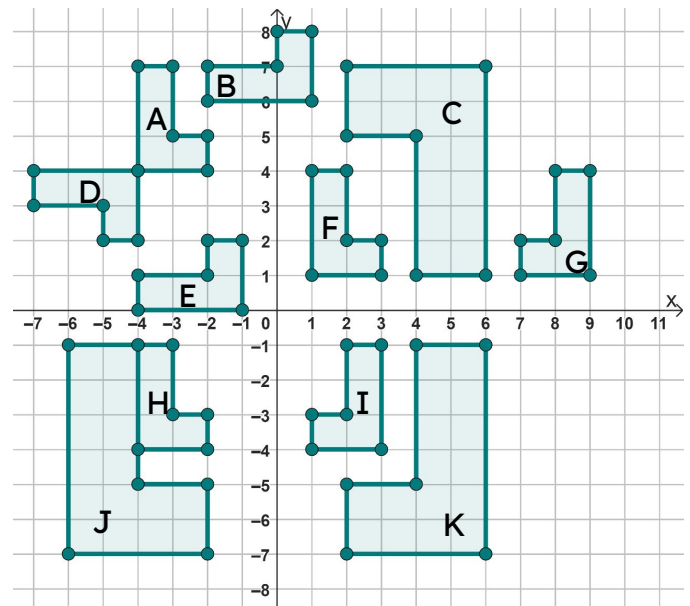
Task B: Recognising and describing transformations

1) Complete the descriptions.

a) Shape A has been reflected in a diagonal line passing through $(-3, 3)$ and $(1, -1)$ onto shape __.

b) Shape __ has been reflected in the y-axis onto shape __.

c) Shape __ has translated by $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$ onto shape __.



d) Shape __ has been enlarged by scale factor 2 from $(-2, -1)$ onto shape __.

e) Shape __ has translated by $\begin{pmatrix} 0 \\ ? \end{pmatrix}$ onto shape A.

f) Shape __ has been rotated 180° about the origin onto shape __.

2) Three different transformations have taken place to create the image, which combined makes a square. What are they?

